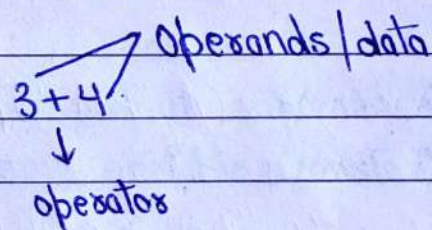


Arithmetic Instruction

An instruction which is used to manipulate data using operators is known as arithmetic



On the basis of operands there are three types of operators -

1. Unary
2. Binary
3. Ternary

Unary operators

$+$, $-$, $++$, $--$, $\text{sizeof}()$

$+$ $\rightarrow$ unary plus

$-$ $\rightarrow$ unary minus

$++$ $\rightarrow$ increment

$--$ $\rightarrow$ decrement

These are of two types -

- | | | |
|-------------------------------|---------------|----------------|
| 1. Prefix increment/decrement | $++x$, $--x$ | first priority |
| 2. Post increment/decrement | $x++$, $x--$ | least priority |

$\text{sizeof}()$

is used to find size of its operand in terms of bytes.
The operand can be variable, constant or any datatype.

Arithmetic operators

$*$ | $\%$

left to Right

$\text{int} / \text{int} = \text{int}$

$\text{float} / \text{int} = \text{float}$

$+$ | $-$

left to Right

$\text{int} / \text{float} = \text{float}$

Ex - $3/4$

$= 0$

$\text{int} / \text{int} = \text{int}$

$\text{real} / \text{int} = \text{real}$

$\text{int} / \text{real} = \text{real}$

$$2.5 \% 3 = \text{error}$$

we cannot apply modulus to real number

$$2 \% 5 = 2$$

5	2	0
	0	
	2	

Bitwise operators

It works on bit (0 and 1) means Bitwise operators operate on integers only.

& AND

| OR

^ XOR

~ NOT

>> Right shift

<< Left shift

$$0 \& 0 \rightarrow 0$$

$$0 \& 1 \rightarrow 0$$

$$1 \& 0 \rightarrow 0$$

$$1 \& 1 \rightarrow 1$$

$$0 | 0 \rightarrow 0$$

$$0 | 1 \rightarrow 1$$

$$1 | 0 \rightarrow 1$$

$$1 | 1 \rightarrow 1$$

$$0 \wedge 0 \rightarrow 0$$

$$0 \wedge 1 \rightarrow 1$$

$$1 \wedge 0 \rightarrow 1$$

$$1 \wedge 1 \rightarrow 0$$

$$\sim 0 \rightarrow 1$$

$$\sim 1 \rightarrow 0$$

Relational operator

< > <= >= left to right
== !=

If relation is true then 1
false then 0

Examples -

$$3 < 4 = 1$$

$$5 > 5 = 0$$

$$2 == 3 = 0$$

$$5 > 4 > 3 = 0$$

{ 5 > 4 True 1 > 3 = 0
1 will print }

logical operators

! NOT & It is a unary operator

& & AND

|| OR

!T = F

!F = T

Assignment operators

$=$ $+=$ $-=$ $*=$ $/=$ $\% =$

Example $\rightarrow$

`int a = 5;`

`a += 2;`

`a = a + 2`

Simple

Simple Interest - $(P * R * T) / 100$

Profit percentage - $((\text{Selling Price} - \text{Cost Price}) / \text{Cost Price}) * 100$

Last digit - $a \% 10$

First digit - $a / 10$, $/ 100$

Middle digit - $(a / 10) \% 10$

Last digit zero - $(a / 10) * 10$

Append number at last - $(a * 10) + \text{Append no.}$